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Partitions based computational method for high-order fuzzy time series forecasting

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ABSTRACT

In this paper, we present a computational method of forecasting based on multiple partitioning and higher order fuzzy time series. The developed computational method provides a better approach to enhance the accuracy in forecasted values. The objective of the present study is to establish the fuzzy logical relations of different order for each forecast. Robustness of the proposed method is also examined in case of external perturbation that causes the fluctuations in time series data. The general suitability of the developed model has been tested by implementing it in forecasting of student enrollments at University of Alabama. Further it has also been implemented in the forecasting the market price of share of State Bank of India (SBI) at Bombay Stock Exchange (BSE), India. In order to show the superiority of the proposed model over few existing models, the results obtained have been compared in terms of mean square and average forecasting errors.

1. Introduction

Relations on the sequential set of past data measured over the time to forecast future values are investigated by time series forecasting. Many statistical tools like regression analysis, moving average, exponential moving average and autoregressive moving average are frequently used in traditional forecasting. One of the biggest challenges of these methods is to address the forecasting problem in which the historical data are in linguistic terms. Zadeh (1965, 1975) introduced the notion of fuzzy set theory to handle the uncertainty and vagueness available in linguistic terms and it was used by Song and Chissom (1993a, 1993b, 1994) and Chen (1996) to develop fuzzy time series models to forecast enrollments of University of Alabama. To enhance the accuracy in forecasted values many researchers, Sullivan and Woodall (1994), Kim and Lee (1999), Hwang, Chen, and Lee (1998), Chen and Hwang (2000), Huarng (2001) proposed various models on fuzzy time series forecasting.

Chen (2002) proposed high-order fuzzy time series modes for the forecasting the enrollments. Dependency between the fuzzy data for the selection of suitable order for fuzzy time series was measured by introducing autocorrelation function by Song (2003). Own and Yu (2005) presented a heuristic higher order model by introducing a heuristic function to incorporate the heuristic knowledge to improve TAIFEX forecast. Cheng, Chang, and Yeh (2006) used trapezoid fuzzification approach and minimize entropy principle approach (MEPA) to partition the universe of discourse improve the fuzzy time series forecasting. Singh (2007a, 2007b, 2008) presented fixed difference parameters based computational algorithm for forecasting with fuzzy time series

Wong, Bai, and Chu (2010) proposed an adaptive time variant forecasting (ATVF) model based on window size of fuzzy time series and tested performance of the proposed method by forecasting the simulated and actual time series including the enrollments of the University of Alabama and the Taiwan Stock Exchange Capitalization Weighted Stock Index (TAIEX). Recently Joshi and Kumar (2012) presented a fuzzy time series model based on non-determinacy index by incorporating intuitionistic fuzzy sets.

To improve the accuracy in the fuzzy time series forecasting, this study proposes a method of forecasting based on nested partitioning of universe of discourse and difference parameters as relations for forecasting. The proposed algorithm minimizes the time of generating relational equations by using complex min-max composition operations and various defuzzification process. The proposed method has been implemented to forecast the historical student enrollments data of University of Alabama and the various other recent methods proposed by Singh (2007a, 2007c), Liu (2007), Cheng, Wang, and Cheng (2008), Wong et al. (2010) and Chen and Tanuwijaya (2011) are selected for benchmarking. Further, it has also been implemented on a real life problem of the movement of market price of share of SBI at BSE, India. Robustness of the proposed model has also been examined by considering the different cases.

The rest of this paper is organised as follows: Section 2 briefly introduces the basic concepts of fuzzy set theory and fuzzy time series. Section 3 proposes and introduces the research framework and proposed computational algorithm. Section 4 verifies the effectiveness of the proposed model with two empirical databases of student enrollments of University of Alabama and market prices

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of SBI share at BSE, India. Finally in Section 5, findings and conclusions are presented.

2. Some basic concepts of fuzzy time series

The various definitions and properties of fuzzy time series are described as follows

Definition 1. Suppose U be the Universe of discourse, $U = \{u_1, u_2, u_3, \ldots, u_n,\}$, where u_k are possible linguistic values of U, then a fuzzy set $\tilde{A}_i$ of U is defined by

$$\tilde{A}_i = \mu_{\tilde{A}_i}(u_1)/u_1 + \mu_{\tilde{A}_i}(u_2)/u_2 + \mu_{\tilde{A}_i}(u_3)/u_3 + \ldots\ldots + \mu_{\tilde{A}_i}(u_n)/u_n$$

where $\mu_{\tilde{A}_i}$ is the membership function of $\tilde{A}_i$, $\mu_{\tilde{A}_i} : U \to [0,1]$ and $\mu_{\tilde{A}_i}(u_k)$ indicates the grade of membership of u_k in $\tilde{A}_i$ where $\mu_{\tilde{A}_i}(u_k) \in [0,1]$ and $1 \leqslant k \leqslant n$.

Definition 2. Suppose $Y(t)$, $(t = \ldots, 0, 1, 2, \ldots)$ be the Universe of discourse and $Y(t) \subseteq R$. Assume that $f_i(t)$, $(i = 1, 2, \ldots)$ is defined in the Universe of discourse $Y(t)$. $F(t)$ is the collection of $f_i(t)$, then $F(t)$ is known as fuzzy time series on $Y(t)$. As at different times, the values of $F(t)$ can be different, $F(t)$ is a function of time and in a similar way, $Y(t)$ may be different at different times.

Definition 3. If $F(t)$ is caused only by $F(t-1)$, denoted by $F(t-1) \to F(t)$, then this relationship can be expressed as the fuzzy relational equation

$$F(t) = F(t-1) \circ R(t, t-1)$$

where the symbol "$\circ$" is Max–Min composition operator. The relation $R(t, t-1)$ is a fuzzy relation between $F(t)$ and $F(t-1)$ and is called the first-order model of $F(t)$.

Definition 4. If $F(t)$ is caused by more fuzzy sets, $F(t-n)$, $F(t-n+1), \ldots, F(t-1)$, the fuzzy relationship is represented by

$$\tilde{A}_{i_1}, \tilde{A}_{i_2}, \tilde{A}_{i_3} \ldots\ldots\ldots\ldots\ldots, \tilde{A}_{i_n} \to \tilde{A}_j$$

here, $F(t-n) = \tilde{A}_{i_1}, F(t-n+1) = \tilde{A}_{i_2}, \ldots, F(t-1) = \tilde{A}_{i_n}$

This relationship is called nth order fuzzy time series model.

Definition 5. Let $F(t)$ be a fuzzy time series and $R(t, t-1)$ be a first order model of $F(t)$. If $R(t, t-1) = R(t-1, t-2)$ for any time t, the $F(t)$ is named a time-invariant fuzzy time series. But if $R(t, t-1)$ is time dependent, that is, $R(t, t-1)$ may be different from $R(t-1, t-2)$ for any time t then $F(t)$ is called time-variant fuzzy time series.

3. Proposed method and algorithm

Section 3.1 briefly introduces the conceptual basis and framework of the proposed model and the steps of the method are described in detail in Section 3.2.

3.1. Proposed method

The method proposes the multiple partitions of time series data using the following expression:

$$\rho = \frac{(E_{\max} + E_{\min})}{2(E_{\max} - E_{\min})} \tag{1}$$

where $E_{\max}$ and $E_{\min}$ are maximum and minimum of time series data, respectively. For each partition, the fuzzy relations and difference parameters are defined by following rules:

(i) For year $n-2$, $n-1$, n, second order fuzzy logical relation is used and difference parameters are computed by $D_i = |E_2 - E_1|$.
(ii) For year $n-3$, $n-2$, $n-1$, n, third order fuzzy logical relation is used and difference parameters are computed by $D_i = ||E_3 - E_2| - |E_2 - E_1||$.
(iii) For year $n-4$, $n-3$, $n-2$, $n-1$, n, fourth order fuzzy logical relation is used and difference parameters are computed by following: $D_i = ||E_4 - E_3| - |E_3 - E_2| - |E_2 - E_1||$.

In a similar manner, the order of fuzzy logical relation is increased for next forecast and the difference parameters are computed accordingly. In general, the following expression is used to compute the difference parameters for each forecast in each partition.

$$D_i = \left| |E_i - E_{i-1}| - \left[\sum_{c=1}^{i-1} |E_{i-c} - E_{i-(c+1)}| \right] + |E_1 - E_0| \right| \tag{2}$$

where $i = 2, 3, \ldots K$ (end of time series data for each partition).

3.2. Computational algorithm of proposed method

This section, presents stepwise computational procedure for forecasting historical time series data of year $n+1$ using the proposed method.

1. Define the Universe of discourse, U based on the range of available time series data, by the rule $U = [E_{min} - E1, E_{max} + E2]$ where $E1$ and $E2$ are two proper positive numbers.
2. Partition the universe of discourse into equal length of intervals: $u1, u2, \ldots, u_m$. The number of intervals will be in accordance with the number of fuzzy sets $\tilde{A}_1, \tilde{A}_2, \ldots, \tilde{A}_m$.
3. Construct the fuzzy sets $\tilde{A}_i$ in accordance with the intervals in Step 2 and apply the triangular membership rule to each interval in each triangular fuzzy set so constructed.
4. Repartition the whole fuzzy time series using the expression given in Section 3.1.
5. Fuzzify the data and establish the fuzzy logical relationships by the rule as discussed in Section 3.1
6. Rules (Computational algorithm) for forecasting

   Some notations used are defined as follows

   $[*\tilde{A}_j]$ is corresponding interval u_j for which membership in $\tilde{A}_j$ is supremum (i.e. 1)
   $L[*\tilde{A}_j]$ is the lower bound of interval u_j
   $U[*\tilde{A}_j]$ is the upper bound of interval u_j
   $M[*\tilde{A}_i]$ is the mid value of the interval u_i having supremum value in $\tilde{A}_i$
   $M[*\tilde{A}_j]$ is the mid value of the interval u_j having supremum value in $\tilde{A}_j$
   For a fuzzy logical relation $\tilde{A}_i \to \tilde{A}_j$
   $\tilde{A}_i$ is the fuzzified enrollment of year n
   $\tilde{A}_j$ is the fuzzified enrollment of year $(n+1)$
   E_i is the actual enrollment of year n
   E_{i-1} is the actual enrollment of year $(n–1)$
   E_{i-2} is the actual enrollment of year $(n–2)$
   E_{i-c} is the actual enrollment of year $(n–c)$
   $E_{i-(c+1)}$ is the actual enrollment of year $\{n–(c+1)\}$
   F_j is the crisp forecasted enrollment of the year $(n+1)$

The proposed method utilizes the historical data of year 1 to n for framing rules to implement on fuzzy logical relation, $\tilde{A}_i \to \tilde{A}_j$, where $\tilde{A}_i$, the current state, is the fuzzified enrollments of year n and $\tilde{A}_j$, the next state, is fuzzified enrollments of year $n+1$. The proposed method for forecasting is mentioned as computational algorithms for generating the relations between the time series data of year 1 to n for forecasting the enrollment of year $n + 1$ in each partition. The developed computational algorithm uses the differences in enrollment of past n years and have been considered a fuzzy parameter in framing the fuzzy rules to impose on current year fuzzified enrollment to get forecast of next year enrollments.

Computational algorithm: Forecasting enrollment for year ($n + 1$) (i.e. 1973 for first partition, 1981 for second partition and 1988 for third partition).

For $k = 1$ to K (Number of partitions)

For $i = 2,3,\ldots N$ (end of time series data of given partition)

Obtained fuzzy logical Relation for year i to $i + 1$

$$\tilde{A}_i \to \tilde{A}_j$$

$$P = 0 \text{ and } Q = 0$$

Compute

$$D_i = \left| |E_i - E_{i-1}| - \left[\sum_{c=1}^{i-1} |E_{i-c} - E_{i-(c+1)}| \right] + |E_1 - E_0| \right|$$

For $a = 2,3,\ldots i$

$F_{ia} = M[*\tilde{A}_i] + D_i/(2(a-1))$

$FF_{ia} = M[*\tilde{A}_i] - Di/(2(a-1))$

If $F_{ia} \geqslant L[*\tilde{A}_j]$ and $F_{ia} \leqslant U[*\tilde{A}_j]$

Then $P = P + F_{ia}$ *and* $Q = Q + 1$

Else $P = P + 0$ *and* $Q = Q + 0$

If $FF_{ia} \geqslant L[*\tilde{A}_j]$ and $FF_{ia} \leqslant U[*\tilde{A}_j]$

Then $P = P + FF_{ia}$ *and* $Q = Q + 1$

Else $P = P + 0$ *and* $Q = Q + 0$

Next a

$F_j = (P + M(*\tilde{A}_j))/(Q + 1)$

Next i

Next k

In fuzzy time series forecasting, mean square error (MSE) and average forecasting error are common tools to measure the accuracy. Lower the MSE or average forecasting error, better the forecasting method. The MSE and average forecasting error are defined as follows:

$$\text{Mean Square Error} = \frac{\sum_{i=1}^{n}(\text{actual value}_i - \text{forecasted value}_i)^2}{n} \tag{3}$$

$$\text{Forecasting Error}(in\%) = \frac{|\text{forecasted value} - \text{actual value}|}{\text{actual value}} \times 100 \tag{4}$$

$$\text{Average forecasting error}(in\%) = \frac{\text{sum of forecasting errors}}{n} \tag{5}$$

4. Proposed model and algorithm

To understand and verification of the proposed model easily, this section implements the method on data of the enrollments at University of Alabama and price of SBI's share at BSE, India.

4.1. Forecasting enrollments with proposed method

Step 1: Universe of discourse $U = [13000, 20000]$.

Step 2: The Universe of discourse is partitioned into seven intervals (linguistic values):

$$u_1 = [13000, 14000], \quad u_2 = [14000, 15000],$$
$$u_3 = [15000, 16000], \quad u_4 = [16000, 17000],$$
$$u_5 = [17000, 18000], \quad u_6 = [18000, 19000],$$
$$u_7 = [19000, 20000].$$

Step 3: Seven fuzzy sets $\tilde{A}_1, \tilde{A}_2, \ldots, \tilde{A}_7$ are define on the universe of discourse U and the membership grades to these fuzzy sets are defined as follows:

$\tilde{A}_1 = 1/u_1 + 0.5/u_2 + 0/u_3 + 0/u_4 + 0/u_5 + 0/u_6 + 0/u_7,$
$\tilde{A}_2 = 0.5/u_1 + 1/u_2 + 0.5/u_3 + 0/u_4 + 0/u_5 + 0/u_6 + 0/u_7,$
$\tilde{A}_3 = 0/u_1 + 0.5/u_2 + 1/u_3 + 0.5/u_4 + 0/u_5 + 0/u_6 + 0/u_7,$
$\tilde{A}_4 = 0/u_1 + 0/u_2 + 0.5/u_3 + 1/u_4 + 0.5/u_5 + 0/u_6 + 0/u_7,$
$\tilde{A}_5 = 0/u_1 + 0/u_2 + 0/u_3 + 0.5/u_4 + 1/u_5 + 0.5/u_6 + 0/u_7,$
$\tilde{A}_6 = 0/u_1 + 0/u_2 + 0/u_3 + 0/u_4 + 0.5/u_5 + 1/u_6 + 0.5/u_7,$
$\tilde{A}_7 = 0/u_1 + 0/u_2 + 0/u_3 + 0/u_4 + 0/u_5 + 0.5/u_6 + 1/u_7.$

Step 4: The fuzzified historical time series data of enrollments are obtained and fuzzy logical relations are established (Table 1).

Step 5: Since $\frac{(E_{\max}+E_{\min})}{2(E_{\max}-E_{\min})} = 2.578$, the time series data again partitioned in three parts. First partition contains enrollments from 1971 to 1978, second partition contains enrollments from 1979 to 1985 and third partition contains enrollments from 1986 to 1992.

Step 6: The computations for the enrollments of University of Alabama have been carried out by using the proposed model (computational algorithm) given in Section 3. The results obtained are placed in the Table 2 along with results of other models.

Mean square error (MSE) and Average Forecasting Error is calculated and are placed in Table 3 to compare the accuracy in forecasted values of proposed model with other models.

The close trend of forecasted enrollments with actual enrollments also confirms the superiority of the proposed method over the methods proposed by Singh (2007a, 2007c), Liu (2007), Cheng et al. (2008), Wong et al. (2010) and Chen and Tanuwijaya (2011) (Fig. 1).

4.2. Robustness of the proposed method

Unexpected perturbation that causes variations in time series data happens very frequently in economic time series. Robustness of any prediction model is its ability to remain stable against such external disturbances. In this section the robustness of the proposed method is tested to show that the proposed method works well even if there is some increment or decrement in enrollments. To see the effect of increase or decrease in the enrollments, years 1973, 1981 and 1988 are selected randomly from each partition. We have considered different cases as follows:

Case 1: If the enrollments in selected years are increased by 5%, MSE is lowered 62976.63 to 61086.19. Average forecasting error is slightly increased from 1.269981 to 1.2811 but is still less than Singh (2007a, 2007c), Liu (2007) and Chen and Tanuwijaya (2011) in the similar situation.

Case 2: If the enrollments in selected years are decreased by 5%, MSE is lowered from 62976.63 to 59030.13 and average forecasting error is almost same.

Case 3: This is more rigorous testing of the model where increment and decrement in the enrollments of selected years take place simultaneously. The following six sub cases are considered in this case.

(i) Increase of 5% in the enrollment of year 1973 and decrease of 5% in the enrollments of years 1981 and 1988.

Table 1
Historical actual enrollments and fuzzified enrollments of University of Alabama

Year	Actual enrollments	Fuzzified enrollments	Year	Actual enrollments	Fuzzified enrollments
1971	13055	$\tilde{A}_1$	1982	15433	$\tilde{A}_3$
1972	13563	$\tilde{A}_1$	1983	15497	$\tilde{A}_3$
1973	13867	$\tilde{A}_1$	1984	15145	$\tilde{A}_3$
1974	14696	$\tilde{A}_2$	1985	15163	$\tilde{A}_3$
1975	15460	$\tilde{A}_3$	1986	15984	$\tilde{A}_3$
1976	15311	$\tilde{A}_3$	1987	16859	$\tilde{A}_4$
1977	15603	$\tilde{A}_3$	1988	18150	$\tilde{A}_6$
1978	15861	$\tilde{A}_3$	1989	18970	$\tilde{A}_6$
1979	16807	$\tilde{A}_4$	1990	19328	$\tilde{A}_7$
1980	16919	$\tilde{A}_4$	1991	19337	$\tilde{A}_7$
1981	16388	$\tilde{A}_4$	1992	18876	$\tilde{A}_6$

(ii) Increase of 5% in the enrollment of year 1981 and decrease of 5% in the enrollments of years 1973 and 1988.
(iii) Increase of 5% in the enrollment of year 1988 and decrease of 5% in the enrollments of years 1973 and 1981.
(iv) Decrease of 5% in the enrollment of year1988 and increase of 5% in the enrollments of years 1973 and 1981.
(v) Decrease of 5% in the enrollment of year 1973 and increase of 5% in the enrollments of years 1981 and 1988.
(vi) Decrease of 5% in the enrollment of year 1981 and increase of 5% in the enrollments of years 1973 and 1988.

Table 4 shows that in all sub cases of case 3, MSE and average forecasting error are lowered compared to actual data set except in the case 3 (v). In case 3(v), MSE and average forecasting errors are increased from 62976.63 to 66733.19 and from 1.269981 to 1.3785 respectively, but still lower than the errors of the models proposed by Singh (2007a, 2007c), Liu (2007) and Chen and Tanuwijaya (2011) in similar situation. This shows the robustness of the proposed method i.e. increasing or decreasing the actual enrollments, the increment or decrement in errors is extremely small. Further, the errors in the methods proposed by Singh (2007a, 2007c), Liu (2007) and Chen and Tanuwijaya (2011) are much higher than our proposed method in similar situations.

4.3. Forecasting market price of SBI's share at Bombay Stock Exchange (BSE)

Because of the nonlinearity of the stock market behaviour, it is extremely hard to forecast accurately without expert knowledge. In this section, we apply the proposed method for forecasting the market price of share of SBI at BSE. The annual reports of State bank of India for year 2008-09 and 2009-10 are used to collect the market price database for SBI share The actual market price of share of SBI for each month from April 2008 to March 2010 is given in Table 5.

Step 1: E_{min} and E_{max} are observed from Table 5 to define the universe of discourse $U = [1000, 2600]$.
Step 2: Partitioned the universe of discourse into eight intervals of linguistic values:

$$u_1 = [1000, 1200] \quad u_2 = [1200, 1400]$$
$$u_3 = [1400, 1600] \quad u_4 = [1600, 1800]$$

$$u_5 = [1800, 2000] \quad u_6 = [2000, 2200]$$
$$u_7 = [2200, 2400] \quad u_8 = [2400, 2600]$$

Table 2
Comparison of enrollments forecast by proposed method and other various methods

Year	Actual	Singh (2007c)	Singh (2007a)	Liu (2007)	Cheng et al. (2008)	Wong et al. (2010)	Chen and Tanuwijaya (2011)	Proposed
1971	13055	–	–	–	–	–	–	–
1972	13563	–	–	13500	14242	–	13512	–
1973	13867	–	–	13800	14242	13500	13998	13500
1974	14696	14750	14500	14700	14242	14500	14658	14500
1975	15460	15750	15358	15600	15474	15500	15341	15500
1976	15311	15500	15500	15400	15474	15466	15501	15500
1977	15603	15500	15500	15750	15474	15392	15501	15500
1978	15861	15500	15500	15400	15474	15549	15501	15500
1979	16807	16500	16500	16800	16147	16433	17065	–
1980	16919	16500	16500	17100	16988	16656	17159	–
1981	16388	16500	16500	17100	16988	16624	17159	16500
1982	15433	15500	15581	15300	16147	15556	15341	15500
1983	15497	15500	15500	15750	15474	15524	15501	15500
1984	15145	15250	15500	15400	15474	15497	15501	15500
1985	15163	15500	15500	15300	15474	15305	15501	15500
1986	15984	15500	15500	15750	15474	15308	15501	–
1987	16859	16500	16402	16800	16147	16402	17065	–
1988	18150	18500	18500	17100	16988	18500	17159	18500
1989	18970	18500	18500	18900	19144	18534	18832	18500
1990	19328	19500	19471	19200	19144	19345	19333	19337
1991	19337	19500	19500	19050	19144	19423	19083	19500
1992	18876	18750	18651	19050	19144	18752	19083	18704

Table 3
Comparison of MSE and average forecasting error of proposed model with other models.

Model	Singh (2007c)	Singh (2007a)	Liu (2007)	Cheng et al. (2008)	Wong et al. (2010)	Chen and Tanuwijaya (2011)	Proposed
MSE	76509	87025	108097	228909.4	88337.2	122085	62976.63
AFE	1.40702	1.55868	1.32783	2.39615	1.51894	1.53526	1.269981

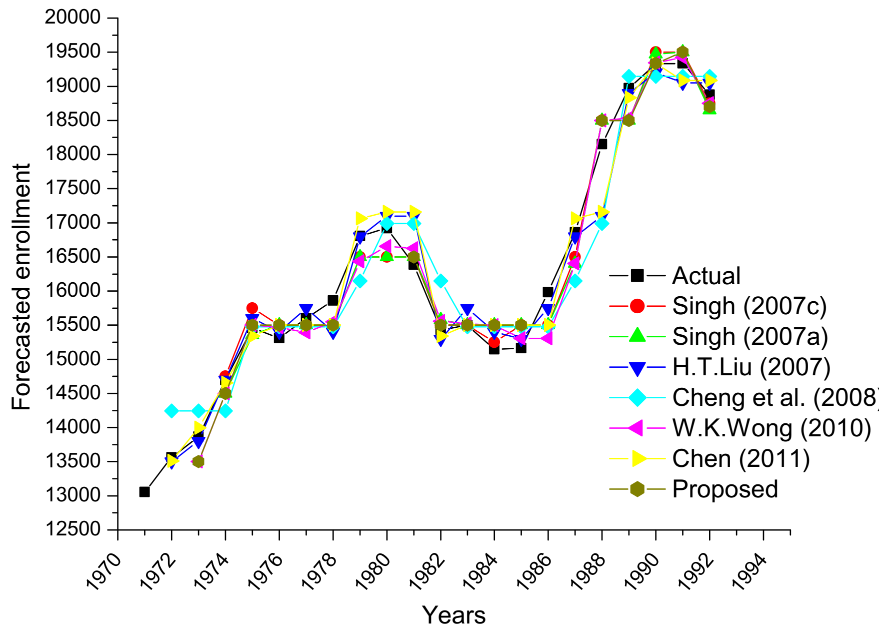

Fig. 1. Actual enrollments vs forecasted enrollments.

Table 4
Comparison of robustness of proposed method with other model under the different situations of fluctuation in actual time series data.

Model	Error	Cases							
		Case 1	Case 2	Case 3					
				(i)	(ii)	(iii)	(iv)	(v)	(vi)
Proposed	MSE	61086.19	59030.13	53383.13	64098.13	61665.19	58451.13	66733.19	56018.19
	AFE	1.2811	1.2696	1.1722	1.3483	1.2998	1.2509	1.3785	1.2024
Singh (2007c)	MSE	101755.2	73786.58	73786.58	78054.37	97487.37	78054.37	101755.2	97487.37
	AFE	1.5592	1.3543	1.3543	1.4207	1.4929	1.4207	1.5592	1.4929
Singh (2007a)	MSE	89836.16	83151.74	83006.53	87659.79	87670.63	85317.26	92178.68	87525.42
	AFE	1.5237	1.4742	1.4637	1.5591	1.4747	1.5232	1.5596	1.4642
Liu (2007)	MSE	172581.8	135304.3	119205.4	365222.8	354276.4	349123.9	188680.6	338177.6
	AFE	1.7135	1.7162	1.5925	2.2251	2.1529	2.1014	1.8372	2.0293
Chen and Tanuwijaya (2011)	MSE	192012	152572	151669.9	380297.6	386829.5	581364.7	206744.9	372118.3
	AFE	1.8391	1.9042	1.8339	2.3023	2.4138	2.9371	1.9873	2.2564

Table 5
Actual enrollments and fuzzified enrollments of market prices of SBI's share.

Year	SBI's Share Price at BSE (Rs.)	Fuzzified enrollments	Year	SBI's Share Price at BSE (Rs.)	Fuzzified enrollments
Apr. 2008	1819.95	$\tilde{A}_5$	Apr. 2009	1355.00	$\tilde{A}_2$
May 2008	1840.00	$\tilde{A}_5$	May 2009	1891.00	$\tilde{A}_5$
Jun. 2008	1496.70	$\tilde{A}_3$	Jun. 2009	1935.00	$\tilde{A}_5$
Jul. 2008	1567.50	$\tilde{A}_3$	Jul. 2009	1840.00	$\tilde{A}_5$
Aug. 2008	1638.90	$\tilde{A}_4$	Aug. 2009	1886.90	$\tilde{A}_5$
Sep. 2008	1618.00	$\tilde{A}_4$	Sep. 2009	2235.00	$\tilde{A}_7$
Oct. 2008	1569.90	$\tilde{A}_3$	Oct. 2009	2500.00	$\tilde{A}_8$
Nov. 2008	1375.00	$\tilde{A}_2$	Nov. 2009	2394.00	$\tilde{A}_7$
Dec. 2008	1325.00	$\tilde{A}_2$	Dec. 2009	2374.75	$\tilde{A}_7$
Jan. 2009	1376.40	$\tilde{A}_2$	Jan. 2010	2315.25	$\tilde{A}_7$
Feb. 2009	1205.90	$\tilde{A}_2$	Feb. 2010	2059.95	$\tilde{A}_6$
Mar. 2009	1132.25	$\tilde{A}_1$	Mar. 2010	2120.05	$\tilde{A}_6$

Step 3: Eight fuzzy sets $\tilde{A}_1, \tilde{A}_2, \ldots, \tilde{A}_8$, are defined on the universe of discourse U and membership grades to these fuzzy sets are defined as follows:

$\tilde{A}_1 = 1/u_1 + 0.5/u_2 + 0/u_3 + 0/u_4 + 0/u_5 + 0/u_6 + 0/u_7 + 0/u_8,$
$\tilde{A}_2 = 0.5/u_1 + 1/u_2 + 0.5/u_3 + 0/u_4 + 0/u_5 + 0/u_6 + 0/u_7 + 0/u_8,$
$\tilde{A}_3 = 0/u_1 + 0.5/u_2 + 1/u_3 + 0.5/u_4 + 0/u_5 + 0/u_6 + 0/u_7 + 0/u_8,$
$\tilde{A}_4 = 0/u_1 + 0/u_2 + 0.5/u_3 + 1/u_4 + 0.5/u_5 + 0/u_6 + 0/u_7 + 0/u_8,$
$\tilde{A}_5 = 0/u_1 + 0/u_2 + 0/u_3 + 0.5/u_4 + 1/u_5 + 0.5/u_6 + 0/u_7 + 0/u_8,$
$\tilde{A}_6 = 0/u_1 + 0/u_2 + 0/u_3 + 0/u_4 + 0.5/u_5 + 1/u_6 + 0.5/u_7 + 0/u_8,$
$\tilde{A}_7 = 0/u_1 + 0/u_2 + 0/u_3 + 0/u_4 + 0/u_5 + 0.5/u_6 + 1/u_7 + 0.5/u_8,$
$\tilde{A}_8 = 0/u_1 + 0/u_2 + 0/u_3 + 0/u_4 + 0/u_5 + 0/u_6 + 0.5/u_7 + 1/u_8.$

Table 6
Comparative study of forecasted enrollments by Chen, Singh and proposed model.

Year	Actual	Chen (1996)	Singh (2007b)	Proposed
Apr. 2008	1819.95	–	–	–
May 2008	1840.00	1900	–	–
Jun. 2008	1496.70	1900	–	1500.00
Jul. 2008	1567.50	1500	1499.58	1500.00
Aug. 2008	1638.90	1500	1665.94	1673.14
Sep. 2008	1618.00	1600	1665.94	1700.00
Oct. 2008	1569.90	1600	1499.58	1512.17
Nov. 2008	1375.00	1500	1300.00	1313.75
Dec. 2008	1325.00	1433	1300.00	1300.00
Jan. 2009	1376.40	1433	1300.00	1300.00
Feb. 2009	1205.90	1433	1300.00	1300.00
Mar. 2009	1132.25	1433	1100.00	1136.06
Apr. 2009	1355.00	1300	1300.00	–
May 2009	1891.00	1433	1900.00	–
Jun. 2009	1935.00	1900	1900.00	1900.00
Jul. 2009	1840.00	1900	1900.00	1900.00
Aug. 2009	1886.90	1900	1900.00	1900.00
Sep. 2009	2235.00	1900	2300.00	2257.03
Oct. 2009	2500.00	2300	2400.00	2493.45
Nov. 2009	2394.00	2300	2300.00	2340.99
Dec. 2009	2374.75	2300	2300.00	2300.00
Jan. 2010	2315.25	2300	2300.00	2300.00
Feb. 2010	2059.95	2300	2100.00	2139.10
Mar. 2010	2120.05	2100	2100.00	2100.00

Step 4: The market prices of SBI's share at BSE are fuzzified with triangular membership function and are placed in Table 5.
Step 5: Since $\frac{(E_{\max}+E_{\min})}{2(E_{\max}-E_{\min})} = 1.3278$, the time series data again partitioned in two parts. First partition contains data from April 2008 to March 2009 and second partition contains data April 2009 to March 2010.
Step 6: The computations have been carried out to forecast the market prices of share of SBI at BSE by using the proposed method (Computational algorithm) given in Section 3. The results obtained are placed in the Table 6 along with forecasted values proposed by Chen (1996) and Singh (2007b).

To examine whether the proposed model has made improvements in forecasting accuracy, two fuzzy time series models, devised by Chen (1996) and Singh (2007b) are used as comparison models. MSE and average forecasting errors are used as evaluation criteria to compare the forecasting performance. Table 7 shows the significant improvements of proposed models over listing models. The forecasted prices obtained by the proposed method are significantly in closed accordance to the actual price (Fig. 2).

Table 7
MSE and average error of market price of SBI's share forecast.

Model	Chen (1996)	Singh (2007b)	Proposed
Mean Square Error (MSE)	35066.98	3608.053	2785.043
Average Forecasting Error	8.266529	3.18418	2.66067

5. Conclusions

In this paper we have proposed a new fuzzy time series method considering order of fuzzy logical relation as parameter in multiple partitioning to enhance the accuracy in forecasting problems. The algorithm of the proposed method is simple as it minimizes the complicated computations of fuzzy relational matrices and avoids the search for a suitable defuzzification process. The proposed method has been tested for forecasting efficiency on the historical time series data of enrollments of University of Alabama and has a comparative study with existence methods. Further the method is also been implemented to forecast the market price of SBI share at BSE, India. From Table 3, we can see that in the forecasting the enrollments at University of Alabama, our proposed method outperforms the method proposed by Singh (2007a, 2007c), Liu (2007), Cheng et al. (2008), Wong et al. (2010) and Chen and Tanuwijaya (2011). Table 4 shows the more robustness of the proposed model than that of Singh (2007a, 2007c), Liu (2007) and Chen and Tanuwijaya (2011). Table 7 shows the outpeformance of our proposed method in forecasting the market price of SBI share at BSE, India compared to the model suggested by Singh (2007b) and Chen (1996). As the work of Singh (2007b) outperformed the work of Chen (1996), Song and Chissom (1993a, 1993b, 1994), Huarng (2001), Lee and Chou (2004) and Tsaur,

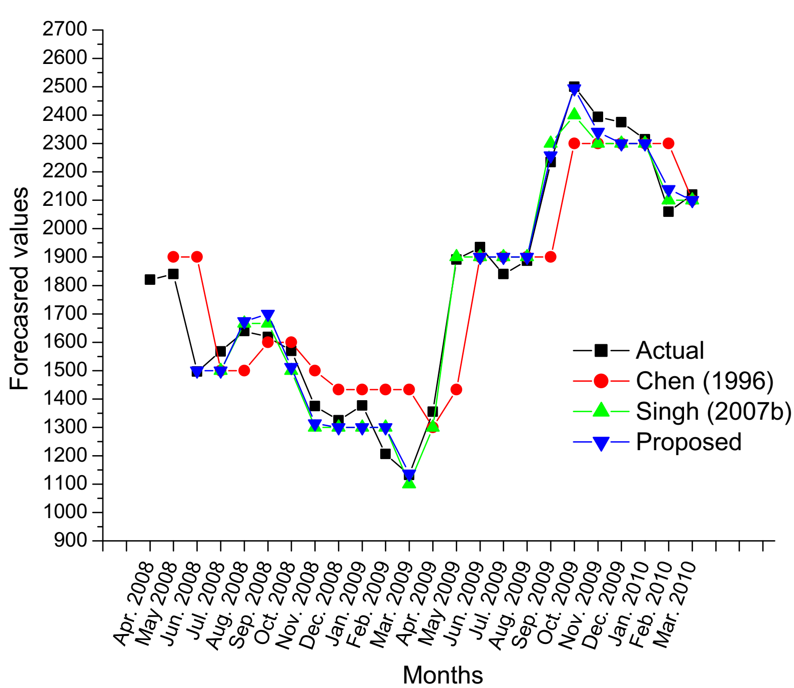

Fig. 2. Actual versus Chen, Singh and Proposed model of SBI's at BSE.

Yang, and Wang (2005), indirectly we can conclude our proposed model outperforms these models in the forecasting the market price of SBI share at BSE, India.

Even though this study has made great improvement in forecasting using fuzzy time series, there are following limitations with proposed method:

1. Time series data are partitioned using the ratio ρ defined by Eq. (1). If $0 < \rho \leqslant 1$, we have one partition. In this case number of difference parameters D_i increased heavily and computation becomes very complicated
2. If $\rho \geqslant \frac{N}{2}$, (where N = no. of time series data), then we will not have sufficient data in each partitions for next forecast.

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